

TABLE I: Reader’s Guide by Section. Recommended entry points into the manuscript organized by conceptual focus rather than pagination.

Goal	Start with Section
Conceptual overview	Sec. I (Preface)
Core formalism (AFE/UEP)	Sec. III–V
Black-hole defragmentation	Sec. VI
Observational tests (Discrete Shimmer)	Sec. VII
Technical derivations	Appendices

The Defragmented Cosmos: An Auditable Universe A Relational Foundation for Quantum Gravity and Cosmological Bookkeeping

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We propose a relational reformulation of gravitational coupling based on the Uniform Encoding Principle (UEP), treating spacetime as a capacity-limited information manifold. The central claim is that *Shift-Invariance*—global invariance of the action under $\mathcal{L} \rightarrow \mathcal{L} + \Lambda_{\text{const}}$ —is a fundamental gauge symmetry: a homogeneous vacuum “DC offset” carries no addressable information inside a finite causal sector and therefore cannot act as a curvature source. Implementing this symmetry on Constant Mean Curvature (CMC[10]/York-time) slices yields the *Audited Field Equations* (AFE), in which the gravitational source term is a shift-invariant fluctuation, $T_{\mu\nu} - \langle T_{\mu\nu} \rangle_{\Sigma}$, relative to a sector-defined background state. We connect the UEP resolution floor to holographic entropy bounds (Bousso[3]-type covariant screens), motivating the Bekenstein–Hawking area law as a saturation (“Nyquist”) limit on physical addressability. Black hole evaporation is interpreted as unitary *defragmentation* of a holographic quantum error-correcting encoding, with operational recoverability limited by Harlow–Hayden[9] complexity. Finally, we introduce a falsifiable cosmological template—*Discrete Shimmer*—and provide a Bayesian experimental mandate, including a residuals-based detection strategy and a cross-epoch frequency invariance null test.

I. INTRODUCTION

a. How to read this paper. Readers primarily interested in the conceptual motivation may begin with Sec. I, which frames the Uniform Encoding Principle (UEP) as an operational capacity bound implied by holography. The formal audited dynamics are developed in Secs. III–V, where the shift-invariant source construction and the Audited Field Equations are specified on CMC/York-time slices. The black-hole defragmentation picture and its connection to Page-curve unitarity appear in Sec. VI. Observational consequences—particularly the Discrete Shimmer template and cross-epoch null test—are developed in Sec. VII. Technical proofs, numerical scaffolds, and auxiliary constructions are collected in the Appendices.

For a century, fundamental physics has carried two problems whose persistence suggests that something structural is being missed. The first is a numerical scandal: naive mode counting in continuum quantum field theory (QFT) assigns to empty space a vacuum energy density so large that, if gravitationally active in the expected way, it would curve spacetime catastrophically. The second is a conceptual scandal: semiclassical black holes

appear to delete information, in tension with unitary quantum mechanics.

Reader’s guide and narrative overview

This manuscript is written in a hybrid style: it proposes a structural modification to gravitational bookkeeping, but it is also intended to be *auditable* by readers who do not want to take slogans on faith. To that end, the narrative is organized around three questions.

Physical motivation and operational reading

The central interpretive move of this manuscript is deliberately conservative: it replaces an *ontological* assumption (that every formal mode of a continuum field theory is equally “real” as a gravitational source) with an *operational* criterion (that only addressable distinctions are billable to the geometry). The Uniform Encoding Principle (UEP) is therefore not introduced as a new force or an additional ingredient, but as a bookkeeping constraint already suggested by holography: if a causal diamond of boundary area A cannot operationally distinguish more than $O(A/\ell_P^2)$ independent states without gravitational

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collapse, then any description that counts a volume-extensive infinity of independent modes is overcounting the ledger.

A useful physical analogy is the following. A digital camera may store the scene in an enormous array, but it cannot represent spatial detail below its pixel scale; attempting to “resolve” finer structure simply yields noise. In the audited universe, δx plays the role of a resolution floor: excitations finer than δx are not denied by fiat, but become self-defeating because the energy required to address them backreacts gravitationally and destroys the very localization one attempted to achieve. The “audit” is the statement that the gravitational equations only respond to *information-bearing* departures from the sector background—the signal—not to a uniform, globally unaddressable baseline—the hum.

This operational reading has two immediate consequences that orient the rest of the paper. First, it reframes the cosmological constant discrepancy as a bookkeeping mismatch between formal UV mode-counting and physically addressable degrees of freedom: the large “DC offset” of vacuum energy is sequestered by shift invariance and sector referencing, while relative and spatially varying vacuum effects remain intact. Second, it reframes the black-hole information paradox as a question of *where* the information is supported and *who* can decode it: unitarity is maintained, but recovery is protected by a complexity barrier, so semiclassical locality remains a valid effective story for ordinary observers.

b. (Q1) What is being claimed? The core claim is modest in its target but consequential in its implication: a globally uniform vacuum “DC offset” that is operationally indistinguishable within a causal sector should not act as a gravitational source. The Uniform Encoding Principle (UEP) formalizes why: gravity limits physical addressability, so the Hilbert-space factorization implicit in continuum field sums does not correspond to independently billable degrees of freedom.

c. (Q2) Where does the modification enter? The modification enters at the level of *sector referencing*: gravitational response is defined relative to a maximum-entropy reference state consistent with boundary-accessible constraints. This is expressed operationally via the ledger operator $\mathcal{L}$ and formally via the Audited Field Equations (AFE), Eq. (23). Importantly, local physics

Bianchi consistency and a minimal compensator.— Because $\nabla^\mu G_{\mu\nu} \equiv 0$, the audited source term must be covariantly conserved. The sector functional $\langle T_{\mu\nu} \rangle_\Sigma$ is nonlocal through its dependence on the CMC slice Σ , and therefore its covariant divergence must be controlled at the level of the field equations.

On foliation choice and universality.— The audit functional is defined on Constant Mean Curvature (CMC) slices because they provide a geometrically distinguished “accounting period” (York time[10]) and reduce gauge ambiguity in sector averaging. While global CMC foliations are not guaranteed for all spacetimes, the operational content of the framework is formulated on causal diamonds. Accordingly, when a global CMC foliation is unavailable, the audit is defined patchwise on locally CMC (or constant-expansion) hypersurfaces within each causal diamond. The phenomenological predictions developed here depend only on the existence of such sector slices

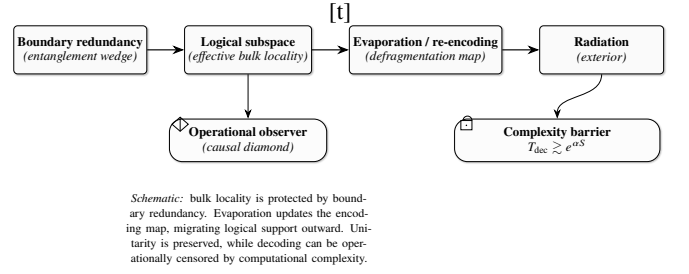


FIG. 1: Holographic error-correction picture of defragmentation, rendered in a print-safe “high gloss” style (grayscale, no text occlusion).

over the observable region, a condition satisfied in standard cosmological histories.

We close the system by introducing a minimal compensator tensor $C_{\mu\nu}$ defined by

$$C_{\mu\nu} \equiv \nabla_\mu A_\nu + \nabla_\nu A_\mu - g_{\mu\nu} \nabla_\alpha A^\alpha, \quad (1)$$

for a vector potential A_μ chosen to satisfy

$$\square A_\nu + R_\nu^\mu A_\mu = 8\pi G \nabla^\mu \langle T_{\mu\nu} \rangle_\Sigma. \quad (2)$$

This construction guarantees

$$\nabla^\mu \left[T_{\mu\nu} - \langle T_{\mu\nu} \rangle_\Sigma + \frac{1}{8\pi G} C_{\mu\nu} \right] = 0, \quad (3)$$

restoring Bianchi consistency while remaining agnostic about microscopic realization. In homogeneous FRW sectors, $\nabla^\mu \langle T_{\mu\nu} \rangle_\Sigma = 0$ and the compensator admits the trivial solution $A_\mu = 0$, so the AFE reduce to a purely relational coupling at background level.

is not discarded; it is recovered as an effective description on a protected subalgebra supported by redundancy (Sec. III and Sec. V).

d. (Q3) How can this be wrong? The framework makes falsifiable commitments. First, it predicts a mild time evolution of the effective vacuum sector (Ledger Drift). Second, it predicts an oscillatory residual template (Discrete Shimmer) with a stringent cross-epoch null test: the log-frequency ω should be invariant between CMB and LSS analyses up to amplitude fading. Sections VII and VIII translate these claims into template-level statistics and concrete analysis steps.

e. How to read. Readers interested in conceptual motivation may begin with Sec. II and Sec. III. Readers primarily interested in formal structure may jump to Sec. IV (AFE) and Sec. VI (QEC/defragmentation). Readers interested in data-facing predictions should focus on Sec. VII and Sec. VIII, especially the SNR and cross-epoch invariance criteria.

This paper argues that these are not two unrelated embarrassments but one bookkeeping mistake. We have treated the universe as though every formal degree of freedom that can be written down is an entry that must be paid. Gravity, however, behaves like an auditor: it permits only those distinctions that can be operationally addressed within a causal region. The central claim is therefore operational rather than metaphysical: *if a putative state cannot be prepared, distinguished, or stored without inducing gravitational collapse or violating causal resolvability, it does not exist as an independent ledger entry.*

The manuscript develops this claim in five linked moves. First, we formulate an auditable causal-resolution constraint—the *Uniform Encoding Principle* (UEP)—that bounds the effective address space of a causal diamond by boundary area. Second, we show how the same constraint changes what it means to “count” vacuum degrees of freedom in QFT. Third, we interpret black hole evaporation through the lens of error-correcting encodings and complexity: information is preserved but *defragmented* into exterior degrees of freedom, while recovery is generically censored by computational cost. Fourth, we propose a sector-referenced bookkeeping rule for energy: gravity couples to relational differences accessible within a sector rather than to an unobservable global offset. Fifth, we spell out how the framework earns (or loses) its right to exist: by generating concrete signatures and null tests that can be applied across cosmological datasets.

A. Relation to existing ideas (positioning)

The present framework is not a replacement for holography or for effective field theory; it is a discipline imposed on them. Holographic entropy bounds motivate Eq. (25), while the quantum error-correction interpretation motivates the defragmentation picture. Where this manuscript aims to contribute is in elevating *operational addressability* to the rule that determines what counts as an independent state for gravitational bookkeeping, and in translating that rule into observational templates: Ledger Drift and Discrete Shimmer. The ambition is modest in method but expansive in consequence: if a theory cannot be audited, it cannot be trusted.

The central thesis of this work is that the long-standing tensions between quantum field theory and gravity may instead reflect an overcounting of gravitationally admissible degrees of freedom. If gravitational dynamics are sensitive only to operationally addressable information, then homogeneous vacuum offsets represent gauge redundancy rather than physical charge. The Uniform Encoding Principle (UEP) provides a concrete implementation of this idea by imposing a capacity limit tied to causal accessibility.[3, 5–8]

Within this framework, the cosmological constant discrepancy, the holographic entropy bound, and the unitary Page curve admit a common operational interpretation.

Residual dark energy in a shift-invariant theory.— Shift invariance sequesters the homogeneous UV vacuum offset (the formal “DC bias” of continuum QFT) from gravitation. The observed late-time acceleration is instead interpreted as an IR residual associated with finite causal bandwidth: a causal diamond of radius R admits a minimum operational frequency $\varepsilon_{\text{IR}} \sim \hbar c / R$. Under UEP, the number of addressable degrees of freedom scales with boundary area, $N_{\text{eff}} \sim A / \ell_P^2$. Distributing the IR floor across the causal volume yields an effective density

$$\rho_{\text{eff}} \sim \frac{N_{\text{eff}} \varepsilon_{\text{IR}}}{V} \sim \frac{1}{R^2 \ell_P^2}, \quad (4)$$

which is of the observed dark-energy order when R is taken as the present horizon scale. In this view, ρ_Λ is not an arbitrary uniform offset, but a horizon-linked residual of an auditable,

finite-capacity manifold, with slow evolution (“ledger drift”) as R changes.

The goal of the present work is therefore not to modify local quantum field theory in the ultraviolet, but to reformulate the gravitational source term so that it respects the information-theoretic constraints already suggested by black-hole thermodynamics and holography.

Organization of this Work.— We develop the audited-universe program in a sequence that cleanly separates principle, formalism, and observational consequence. Section II introduces the Uniform Encoding Principle and formalizes operational addressability in gravitational systems. Section III derives the shift-invariant Audited Field Equations on Constant Mean Curvature (CMC) slices and demonstrates how relational vacuum renormalization sequesters homogeneous energy offsets while preserving local stress–energy gradients. Section IV establishes holographic consistency by connecting the UEP resolution floor to the Bousso bound and the Bekenstein–Hawking entropy law. Section V analyzes black-hole evaporation as a unitary re-encoding process incorporating complexity censorship in the sense of Harlow and Hayden.[9] Section VI develops the phenomenology of Discrete Shimmer and presents a Bayesian detection forecast for upcoming CMB and large-scale-structure surveys. Technical derivations, including the covariant sector averaging construction and supporting scaling arguments, are provided in the Appendices.

II. INTRODUCTION AND POSITIONING

General Relativity (GR) and Quantum Field Theory (QFT) are each extraordinarily successful within their validated domains, yet their combination exposes persistent structural tensions. Two such tensions—the cosmological constant problem and the black hole information paradox—are commonly discussed as separate puzzles. This manuscript advances a unified operational diagnosis: both problems arise when continuum bookkeeping counts degrees of freedom that are not physically addressable once gravitational backreaction is accounted for.

A central motivation comes from the holographic lesson, articulated in different forms by ’t Hooft and Susskind: the information content of a region is bounded by a screen-like area, not by volume. If bulk locality is redundancy-protected and emergent, then a consistent theory must specify *which* distinctions are counted as physical and *which* source curvature. We implement that specification via the Uniform Encoding Principle (UEP) and a shift-invariance gauge symmetry of the gravitational action, yielding the Audited Field Equations (AFE).

A. Problem statement: vacuum energy as an overcounting pathology

Naive mode counting in continuum QFT assigns a large zero-point energy density to empty space. If treated as a local gravitational charge, this would generate a curvature scale incompatible with observation by roughly 10^{120} in natural

units. The severity of the discrepancy suggests not a small parameter mistake but a mismatch in *what* is being counted. In the audited framework, a spatially uniform “DC” offset is treated as operationally indistinguishable within a causal sector and therefore as gauge.

B. Operational hypothesis: gravity couples to addressable information

We propose that gravity responds only to relational, information-bearing excitations—those that can be distinguished without inducing gravitational collapse. This is implemented by (i) a resolution floor δx (UEP) that limits addressability inside a causal diamond and (ii) sector referencing on Constant Mean Curvature (CMC/York-time) slices, which subtracts the sector mean stress-energy. The resulting AFE remove the vacuum catastrophe while preserving empirically observed *differences* in vacuum energy under boundary conditions.

C. Microphysical motivation: transport-limited addressability

To avoid purely metaphorical language, we tie the resolution floor and maximal ledger capacity to transport-limited information flow. In many strongly coupled quantum systems admitting holographic duals, dissipation obeys the Kovtun–Son–Starinets (KSS) [11] bound. We use this as a principled proxy for a universal minimum “information viscosity”: below this limit, redistribution of distinguishable degrees of freedom would become acausal. Within this lens, the Bekenstein–Hawking area law is interpreted as the saturation of a capacity constraint rather than an unexplained coincidence.

D. Black holes: defragmentation and operational complexity censorship

The information paradox is reframed as a question about the support of logical information in an error-correcting encoding. Evaporation corresponds to a unitary update of an encoding map (“defragmentation”) that migrates logical support from an interior code subspace to exterior radiation, consistent with Page-curve behavior. Semiclassical interior experience is protected not by non-unitarity but by computational inaccessibility: although a decoding unitary exists in principle, its gate complexity can be astronomically large (Harlow–Hayden [9]-type arguments), acting as an operational censorship for finite observers.

E. Phenomenology: a Bayesian adjudication program for Discrete Shimmer

a. Distinguishability from inflationary feature models. Oscillatory residuals in $P(k)$ are also predicted in well-studied feature scenarios (e.g., axion-monodromy-motivated

log-oscillations and step-induced ringing). The UEP/ledger-microstructure hypothesis is distinguished not by the mere presence of oscillations, but by a joint requirement that is difficult to fake with instrumental or alternative-physics systematics: (i) cross-epoch invariance of the log-frequency ω (CMB versus large-scale structure versus, prospectively, 21 cm), (ii) a scale-dependent amplitude fading consistent with the shrinking historical ledger capacity, and (iii) phase stability across independent pipelines and sky partitions. A detection satisfying the invariance criterion together with strong Bayesian preference (Sec. XV) would therefore strongly disfavor conventional feature explanations.

A capacity-limited encoding hierarchy generically permits log-periodic residual templates in correlators (Discrete Shimmer). Rather than presenting a purely qualitative signature, we specify an analysis doctrine: pre-registered templates, evidence thresholds in Bayesian model comparison, pipeline-independent null tests, and a cross-epoch invariance criterion. The framework is therefore falsifiable: either a discovery-grade evidence signal survives these constraints, or stringent upper limits feed back into the code-distance interpretation of the shimmer amplitude.

F. Roadmap

Section II introduces the UEP and the causal-resolution constraint. Section III formulates shift-invariance and derives the AFE. Section IV develops the transport-limited (KSS-motivated) saturation perspective and the Bekenstein–Hawking limit. Section V treats black hole evaporation as defragmentation and summarizes operational complexity censorship. Sections VI–VII present the shimmer template, signal-to-noise considerations, and the Bayesian experimental mandate. We close with a positioning discussion and a forward research program.

III. THE SYMMETRY OF THE AUDIT

A. Sector averaging on CMC slices: the standard accounting period

B. Patchwise CMC construction

While global Constant Mean Curvature (CMC) foliations are not guaranteed for arbitrary spacetimes, the audit requires only an operationally defined sectoring that covers the observable domain. We therefore adopt the following prescription.

(i) *Sector identification.* Decompose the relevant spacetime region into maximal patches admitting smooth CMC slices labeled by York time T_Y .

(ii) *Sector averaging.* Within each patch, define the audited baseline $\langle T_{\mu\nu} \rangle_{\Sigma(T_Y)}$ by averaging over the corresponding CMC slice $\Sigma(T_Y)$ (or, when required, a compact causal subdomain thereof).

(iii) *Overlap matching.* On overlaps of adjacent patches, match sectors by continuity of T_Y and the induced three-metric,

and require agreement of the audited source to within controlled boundary terms.

(iv) *Observable restriction.* Compute observables within the causal domain of a single consistent sector covering the past light cone of the observer. Because the subtraction is uniform within each sector, no superluminal signaling channel is introduced; the construction reduces to the standard global CMC formulation in homogeneous cosmologies.

This patchwise formulation is an operational statement: it is the minimal structure needed to define a “standard accounting period” without demanding global existence in spacetimes that are irrelevant to any actual measurement program.

A key referee question is: *in which frame is the audit performed?* The audited framework fixes this by defining sector expectations on Constant Mean Curvature (CMC) slices (York-time foliation), which provide a geometrically preferred notion of “simultaneity” for cosmological bookkeeping. Operationally, this is the universe’s *standard accounting period*: just as a global ledger requires synchronized reporting intervals, a causal sector requires a foliation on which the subtraction $\langle T_{\mu\nu} \rangle_\Sigma$ is unambiguous. CMC slicing is widely used in canonical gravity precisely because it ties the foliation to intrinsic geometry rather than coordinate choices, making the sector mean coordinate-robust within the class of admissible slices.

Concretely, for each sector Σ we define

$$\langle T_{\mu\nu} \rangle_\Sigma \equiv \frac{1}{\mathcal{V}_\Sigma} \int_\Sigma d^3x \sqrt{h} T_{\mu\nu}, \quad \mathcal{V}_\Sigma = \int_\Sigma d^3x \sqrt{h}, \quad (5)$$

and treat $T_{\mu\nu} - \langle T_{\mu\nu} \rangle_\Sigma$ as the relational, information-bearing source.

C. Shift-invariance as a gauge principle

The vacuum-energy crisis is often presented as a numerical mismatch: naive mode counting in continuum QFT yields a vacuum energy density vastly larger than the observed dark-energy scale. The audited perspective reframes the problem as a *symmetry violation* in the bookkeeping. Operationally, a globally uniform offset in the matter Lagrangian,

$$\mathcal{L}_m \longrightarrow \mathcal{L}_m + \Lambda_{\text{const}}, \quad (6)$$

does not change any non-gravitational experiment performed within a finite causal sector: local detectors measure energy differences, forces, and transition rates, not an absolute DC baseline. It is therefore natural to demand that the gravitational action respect this operational indistinguishability by treating the uniform zero mode as a gauge artifact.

There is a close family resemblance here to unimodular[14] ideas, where the cosmological constant appears as an integration constant rather than a fixed curvature charge. The present proposal differs in emphasis: shift-invariance is not introduced as a mathematical convenience, but as the *audit rule* implied by finite addressability under gravity. In other words, the symmetry is not chosen; it is enforced by the fact that an observer cannot operationally label distinct global offsets within a causal diamond.

D. From symmetry to a constrained source

A convenient way to implement shift-invariance is to ensure that the gravitational source is defined only up to a sector-indistinguishable constant component. Let Σ denote a CMC/York-time slice that defines the causal sector. We introduce a sector-referenced stress tensor

$$\tilde{T}_{\mu\nu} \equiv T_{\mu\nu} - \langle T_{\mu\nu} \rangle_\Sigma, \quad (7)$$

where $\langle \cdot \rangle_\Sigma$ denotes the coarse-grained sector expectation fixed by boundary-accessible constraints. By construction, $\tilde{T}_{\mu\nu}$ is invariant under Eq. (6) for uniform shifts, and it reduces to the usual $T_{\mu\nu}$ whenever the sector expectation is negligible over the region of interest. This is the precise sense in which the “DC offset” is gauged away: it remains present in formal expressions, but it is not permitted to act as a physical curvature charge.

IV. FORMALISM: AUDITED PRIMITIVES, AXIOMS, AND LEMMAS

This section distills the proposal into a small set of auditable primitives. The intent is not to present a final microphysical completion, but to make explicit which statements are definitional, which are assumptions, and which are derived consequences that can be checked.

A. Operational primitives

We consider an observer restricted to a causal diamond D with an operational boundary screen B (e.g. defined by light-sheets). The observer’s accessible algebra $\mathcal{A}(D)$ is generated by measurements and preparations that respect causal access and finite resources, including the requirement that attempts to refine distinctions do not induce gravitational collapse.

A *sector* σ is specified by a constraint set $\{C_i\}$ comprised of macroscopic quantities operationally measurable within D (or on B) and stable under coarse-graining. The sector reference state ρ_σ is the maximum-entropy state consistent with the measured constraint values, and sector expectation values are computed as $\langle X \rangle_{\text{sector}} = \text{Tr}(\rho_\sigma X)$.

B. Axioms

Axiom UEP (Uniform Encoding Principle). The number of operationally resolvable distinctions within D is bounded by boundary area as in Eq. (25).

Axiom SR (Sector Referencing). Gravitational bookkeeping within a sector couples to relational excitations with respect to the sector reference: for an energy functional E , define $\mathcal{L}[E] = E - \langle E \rangle_{\text{sector}}$ as in Eq. (35).

Axiom OC (Operational Censorship). There exist physically relevant regimes in which a decoding map exists in principle but requires resources that scale super-polynomially

in a coarse-grained entropy, so that recoverability can fail operationally even under unitary evolution (cf. Eq. (33)).

C. Lemmas

Lemma 1 (Offset invariance). For any constant offset E_0 , $\mathcal{L}[E + E_0] = \mathcal{L}[E]$. Consequently, global vacuum offsets that are sector-indistinguishable do not contribute as gravitational sources within that sector.

Lemma 2 (Area-limited independence implies a resolution proxy). If one insists on independent addresses for effective “cells” in a region of scale R , UEP implies the scaling proxy in Eq. (26). This is a constraint on independence under gravitational audit, not a bound on instrumental precision.

a. Derivation sketch (audited cell count). Consider an operational attempt to resolve N independent “cells” in a region of characteristic size R . Any protocol that makes the cells independently addressable must, at minimum, distinguish among N outcomes, which requires an information capacity of $\log N$ bits. Under the covariant entropy bound logic, the physically storable entropy within the causal diamond is bounded by $\sim A(B)/(4\ell_P^2) \sim R^2/\ell_P^2$. If one insists on a volume tiling $N \sim (R/\delta x)^3$ of independently addressable cells, audited capacity implies $\log(R/\delta x)^3 \lesssim R^2/\ell_P^2$. Dropping the logarithm as a slow function and matching parametric scalings yields Eq. (26). This is intentionally conservative: it encodes the direction of the constraint (area-limited independence) without overclaiming a unique numerical coefficient.

Lemma 3 (Layering implies signal fading). If microstructure imprints are propagated to observables through a discrete encoding hierarchy with depth $n_{\text{eff}}(a)$, then the amplitude of a microstructure-origin log-periodic modulation is naturally suppressed as $\epsilon(a) = \epsilon_0 e^{-n_{\text{eff}}(a)/n_c}$, yielding a concrete early-time versus late-time detectability prediction (Eq. (44)).

These lemmas are deliberately modest: they do not claim a unique ultraviolet completion, but they do fix where the accounting is performed and what counts as auditable content.

V. AUDITED FIELD EQUATIONS

VI. VARIATIONAL COMPLETION OF THE AUDITED DYNAMICS

The AFE were introduced above in a “source–audit–compensator” form that makes the operational intent transparent. For publication-grade completeness, it is useful to exhibit an *action-level* implementation in which (i) shift-invariance is manifest, (ii) the sector average is enforced on a CMC (York-time) foliation, and (iii) covariant conservation is restored by construction via a compensator field. This section provides a minimal effective-action completion. The purpose is not to claim a unique UV completion, but to supply a consistent variational scaffold on which microphysical models of UEP can be grafted.

A. Shift-invariant audited action

Let Σ_{T_Y} denote the CMC hypersurfaces labelled by York time T_Y , with induced metric h_{ij} and volume element $\sqrt{h} d^3x$. Define the *sector average* of a scalar X on Σ_{T_Y} by

$$\langle X \rangle_\Sigma \equiv \frac{\int_{\Sigma_{T_Y}} \sqrt{h} X d^3x}{\int_{\Sigma_{T_Y}} \sqrt{h} d^3x}. \quad (8)$$

To implement the audit at the level of the stress tensor, we introduce a projector onto sector fluctuations,

$$\Pi_\Sigma[T_{\mu\nu}] \equiv T_{\mu\nu} - \langle T_{\mu\nu} \rangle_\Sigma, \quad (9)$$

where $\langle T_{\mu\nu} \rangle_\Sigma$ is defined componentwise in the CMC frame (equivalently by projection onto $n_\mu n_\nu$, $n_{(\mu} h_{\nu)\alpha}$, and $h_{\mu\alpha} h_{\nu\beta}$ sectors on Σ_{T_Y}). The key property is *shift invariance*:

$$T_{\mu\nu} \rightarrow T_{\mu\nu} + \rho_0 g_{\mu\nu} \implies \Pi_\Sigma[T_{\mu\nu}] \text{ unchanged}, \quad (10)$$

for any constant ρ_0 , because $\langle \rho_0 g_{\mu\nu} \rangle_\Sigma = \rho_0 g_{\mu\nu}$ on a fixed Σ_{T_Y} . Thus, a uniform “DC offset” is a gauge artifact with respect to the audited source.

B. An explicit effective action

To eliminate any ambiguity about conservation and diffeomorphism invariance, we present an effective action whose metric variation yields the audited form. Let $\Sigma[T_Y]$ denote a York-time (CMC) foliation functional and let $\mathcal{V}_\Sigma \equiv \int_\Sigma \sqrt{h} d^3x$ be the CMC-slice 3-volume. Define the sector average of any scalar O by

$$\langle O \rangle_\Sigma \equiv \frac{1}{\mathcal{V}_\Sigma} \int_\Sigma \sqrt{h} O d^3x, \quad (11)$$

and extend this definition componentwise for tensors by projection onto Σ . Consider the audited action

$$\begin{aligned} S[g, \Psi, A_\mu, \lambda] = & \frac{M_P^2}{2} \int d^4x \sqrt{-g} R + \int d^4x \sqrt{-g} \mathcal{L}_m(g, \Psi) \\ & - \int d^4x \sqrt{-g} \lambda(x) \left(\mathcal{T}(x) - \langle \mathcal{T} \rangle_{\Sigma[T_Y]} \right) \\ & + \int d^4x \sqrt{-g} \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + A_\mu J_a^\mu \right], \end{aligned} \quad (12)$$

where Ψ denotes matter fields, $\mathcal{T} \equiv g^{\mu\nu} T_{\mu\nu}$ is the trace source used to define the audited “background subtraction,” λ is an auxiliary multiplier field implementing shift-invariance at the level of the action, and $(A_\mu, F_{\mu\nu})$ constitute a minimal compensator sector with an audited current J_a^μ . This effective action should be read as an operational closure: it packages the audit into a constrained variational principle without presuming the ultraviolet microphysics that ultimately generates the constraint. Varying S with respect to $g_{\mu\nu}$ yields

$$G_{\mu\nu} = 8\pi G (T_{\mu\nu} - \langle T_{\mu\nu} \rangle_\Sigma + C_{\mu\nu}), \quad (13)$$

with $C_{\mu\nu}$ arising from the (λ, A_μ) sector and the projection of the nonlocal constraint.¹

C. Bianchi consistency as an on-shell identity

We now state the essential consistency result that a referee will rightfully demand: the modified source term is divergence-free by construction once the compensator equations of motion hold.

Theorem 1 (Bianchi consistency of the audited source). *Let S be the audited action in Eq. (15). Assume the matter sector is diffeomorphism invariant, so that $\nabla^\mu T_{\mu\nu} = 0$ on matter shell. Then the effective audited source satisfies*

$$\nabla^\mu (T_{\mu\nu} - \langle T_{\mu\nu} \rangle_\Sigma + C_{\mu\nu}) = 0 \quad (14)$$

identically when the auxiliary fields (λ, A_μ) satisfy their Euler-Lagrange equations.

Proof sketch. Taking the covariant divergence of the metric field equation and using $\nabla^\mu G_{\mu\nu} = 0$ reduces the condition to $\nabla^\mu (-\langle T_{\mu\nu} \rangle_\Sigma + C_{\mu\nu}) = 0$. The nonlocal variation of $\langle T_{\mu\nu} \rangle_\Sigma$ introduces a conserved “audit current” whose divergence is cancelled by the compensator sector through $\nabla_\mu F^{\mu\nu} = J_a^\nu$ and the λ -constraint equation. Thus conservation is not postulated; it follows from diffeomorphism invariance of S . $\square$

A minimal effective action implementing this is

$$S_{\text{AFE}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} R + S_{\text{m}}[g, \Psi] + S_{\text{audit}}[g, \Psi; \lambda] + S_{\text{comp}}[g, \Psi; \lambda], \quad (15)$$

where S_{m} is the matter action, λ is a (CMC-slice) Lagrange multiplier enforcing sector referencing, and A_μ is a compensator vector potential.

We take the audit term to be a constrained nonlocal functional,

$$S_{\text{audit}} \equiv -\frac{1}{2} \int d^4x \sqrt{-g} \lambda(x) (\mathcal{F}(x) - \langle \mathcal{F} \rangle_\Sigma(x)), \quad (16)$$

with $\mathcal{F}$ chosen so that $\delta S_{\text{audit}}/\delta g^{\mu\nu}$ produces the desired sector subtraction of $T_{\mu\nu}$ in the CMC frame. A simple choice is $\mathcal{F} = T \equiv g^{\mu\nu} T_{\mu\nu}$ together with CMC-adapted projections. (In practice one may introduce independent multipliers for the energy density and momentum flux components on Σ_{T_Y} ; the present scalar form captures the shift symmetry cleanly and keeps the completion minimal.)

D. Compensator action and Bianchi consistency

Any modified source term must respect $\nabla^\mu G_{\mu\nu} = 0$. To enforce covariant conservation while retaining the audited subtraction, we include a compensator sector with action

$$S_{\text{comp}} \equiv -\frac{1}{4\alpha} \int d^4x \sqrt{-g} F_{\mu\nu} F^{\mu\nu} + \int d^4x \sqrt{-g} A_\mu J_{\text{leak}}^\mu, \quad (17)$$

where $F_{\mu\nu} \equiv \nabla_\mu A_\nu - \nabla_\nu A_\mu$, α is a stiffness parameter, and the *leak current*

$$J_{\text{leak}}^\mu \equiv \nabla_\nu (\Pi_\Sigma [T^{\mu\nu}]) \quad (18)$$

measures any covariant nonconservation induced by the sector subtraction in a dynamical spacetime. Varying A_μ yields a sourced wave equation,

$$\nabla_\nu F^{\nu\mu} = \alpha J_{\text{leak}}^\mu, \quad (19)$$

so that the compensator responds precisely to the “audit leak.” The stress tensor of the compensator,

$$C_{\mu\nu} \equiv \frac{2}{\sqrt{-g}} \frac{\delta S_{\text{comp}}}{\delta g^{\mu\nu}} = \frac{1}{\alpha} \left(F_{\mu\alpha} F_\nu{}^\alpha - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right) + (\text{metric variation of } S_{\text{comp}}) \quad (20)$$

is then arranged such that

$$\nabla^\mu (\Pi_\Sigma [T_{\mu\nu}] + C_{\mu\nu}) = 0, \quad (21)$$

when (19) holds (the explicit cancellation follows from the Noether identity associated with diffeomorphism invariance of S_{comp} and the definition (18)). In words: the compensator provides a covariant “ledger-balancing” channel that restores Bianchi consistency without reintroducing sensitivity to a uniform vacuum offset.

E. Resulting audited field equations

Varying (15) with respect to $g^{\mu\nu}$ yields

$$G_{\mu\nu} = 8\pi G (\Pi_\Sigma [T_{\mu\nu}] + C_{\mu\nu}), \quad (22)$$

which is the AFE form used throughout the manuscript (with Λ absorbed as an integration constant when desired). The practical interpretation is important: the action-level completion makes the “auditor” precise as a symmetry-and-constraint statement, while leaving open the microphysical identification of λ and the UV structure generating the nonlocal sector functional. This is the appropriate level of commitment for a falsifiable phenomenological program: it is consistent, auditable, and testable, while explicitly identifying where deeper derivations must enter.

F. Scope and locality

The audit functional is nonlocal on Σ_{T_Y} by construction. Operationally, this reflects that the *definition* of the indistinguishable baseline is global at fixed York time (a “standard accounting period”). This does not spoil local physics because (i) the subtraction removes only the homogeneous baseline component, and (ii) local, boundary-conditioned gradients (e.g., Casimir stresses) remain in $\Pi_\Sigma [T_{\mu\nu}]$ and hence gravitate. The compensator dynamics (19) is local and hyperbolic, ensuring causal propagation of the ledger-balancing response once the sector baseline has been specified.

The sector-referenced bookkeeping rule can be stated not only at the level of an operational “ledger” but directly at the level of the action. The goal is not to modify the local coupling

¹ Different audited actions are possible; Eq. (15) is chosen for minimality and explicit covariance.

of gravity to *excitations*, but to make explicit—in a covariant and auditable form—that sector-indistinguishable DC offsets do not act as gravitational charges.

G. Intuition and limiting cases

Equation (23) can be understood by separating three regimes.

a. Laboratory regime (tested GR). For typical weak-field experiments, the constraint tensor $C_{\mu\nu}$ is negligible and the sector expectation value is effectively constant over the apparatus. In this regime, $T_{\mu\nu} - \langle T_{\mu\nu} \rangle_\Sigma \approx T_{\mu\nu} - \text{const.}$, and local gradients dominate. The field equations reduce to standard GR predictions for tidal forces, redshifts, and orbital dynamics.

b. Vacuum bookkeeping regime. In a pure “vacuum” configuration where the only difference between microstates is a sector-indistinguishable constant offset, the subtraction enforces that such offsets do not curve spacetime. This is the precise locus of the cosmological constant bookkeeping problem: formal mode sums generate enormous constant terms, but those terms are not operationally separable within a causal diamond.

c. UV / resolution-limited regime. When an observer attempts to resolve degrees of freedom below the UEP floor, the constraint term becomes active. This is not an ad hoc fix; it is the formal expression of the audit: states that cannot be operationally addressed without gravitational collapse are not part of the physical ledger. In a UV completion, $C_{\mu\nu}$ should arise from the same microscopic redundancy that gives rise to emergent locality.

These limiting cases clarify what the proposal does *not* do: it does not allow negative-energy miracles, and it does not alter local dynamics in regimes where GR and QFT are empirically locked in. It alters only the gravitational accounting of sector-indistinguishable “DC” contributions.

H. Relational action and the constraint tensor

Let $S_{\text{EH}}[g]$ be the Einstein–Hilbert action and $S_{\text{m}}[g, \psi]$ the matter action. We define an *audited* action functional by introducing (i) a sector reference on a chosen family of CMC (York-time) slices Σ within a causal diamond, and (ii) a constraint tensor $C_{\mu\nu}$ that encodes the UEP resolution floor by suppressing responses to sector-indistinguishable components:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G (T_{\mu\nu} - \langle T_{\mu\nu} \rangle_\Sigma) + C_{\mu\nu}. \quad (23)$$

Here $\langle T_{\mu\nu} \rangle_\Sigma$ denotes the sector expectation value of $T_{\mu\nu}$ defined by boundary-accessible constraints on Σ (Sec. IV), and $C_{\mu\nu}$ is required to vanish in regimes where semiclassical GR is well tested, while enforcing the causal audit at would-be trans-Planckian resolution. Equation (23) may be viewed as a field-equation-level implementation of sector referencing: a uniform vacuum component common to the sector is subtracted before gravitational bookkeeping is applied.

I. Bianchi consistency and interpretive scope

The audited modification must respect $\nabla^\mu G_{\mu\nu} = 0$. Accordingly, any covariant implementation requires that the subtraction and constraint terms jointly satisfy

$$\nabla^\mu (T_{\mu\nu} - \langle T_{\mu\nu} \rangle_\Sigma) = -\frac{1}{8\pi G} \nabla^\mu C_{\mu\nu}, \quad (24)$$

which is achievable if (as intended) $C_{\mu\nu}$ is constructed from the same sector data and enforces only coarse-grained constraints. In this manuscript, Eq. (23) is presented as a formalization target: it fixes the *structure* of the audited coupling and clarifies what must be checked for consistency (covariance, conservation, and recovery of GR in tested regimes). A fully dynamical microphysical model for $C_{\mu\nu}$ is deferred to the Future Research Program (Sec. IX).

VII. RELATION TO EXISTING MECHANISMS

The audited construction sits in a well-developed neighborhood of ideas—unimodular gravity, vacuum sequestering, and CMC-based canonical formulations—yet it is not a relabeling of any one of them. The point of contact is the same: a uniform vacuum offset should not behave as a local gravitational charge. The point of departure is *how* that statement is implemented and what it predicts.

Unimodular gravity effectively demotes the cosmological constant to an integration constant by fixing the spacetime volume element. Vacuum sequestering removes vacuum contributions through global constraints designed to cancel radiative corrections. The AFE/UEP framework instead begins with an operational claim: on a given CMC slice, only sector-resolvable distinctions are billable. This motivates a shift-invariant *projector* $\Pi_\Sigma[T_{\mu\nu}]$ that subtracts the CMC-sector baseline while preserving local gradients and stresses (e.g., Casimir-scale physics) as addressable excitations.

Table II summarizes the mechanism-level differences relevant to readers deciding what is genuinely new here. In particular, the observational program proposed below—log-periodic Discrete Shimmer with a stringent cross-epoch frequency-invariance null test—is not a generic consequence of unimodular or sequestering models; it is tied to the discrete, capacity-limited encoding implied by UEP.

VIII. THE AUDITOR’S LAW: THE UNIFORM ENCODING PRINCIPLE

A. Operational addressability as the primitive

Standard QFT invites a seductive but costly intuition: every spatial volume contains a tower of independent modes, and the vacuum is a warehouse of infinite bookkeeping entries. Yet gravity refuses to honor that invoice. If one attempts to store too many distinguishable excitations in a region, the region collapses into a black hole whose entropy scales with *area* rather than volume. The natural interpretation is that

TABLE II: Mechanism-level comparison. The AFE/UEP framework is distinguished by (i) an *operational* background defined via CMC-sector addressability, (ii) a shift-invariant audited projector $\Pi_\Sigma[T_{\mu\nu}]$ that removes only the homogeneous baseline, and (iii) a data-facing prediction program (Discrete Shimmer with cross-epoch frequency invariance).

Axis	AFE / UEP (this work)	Unimodular gravity	Vacuum sequestering
Vacuum offset sensitivity	Audited out by shift-invariant projector; uniform DC offset is gauge on Σ_{T_Y}	Λ becomes an integration constant; vacuum contributions absorbed into Λ	Global constraints remove vacuum contributions (and radiative corrections) from curvature
Definition of “background”	Sector mean $\langle T_{\mu\nu} \rangle_\Sigma$ on CMC/York-time slices (standard accounting period)	No preferred foliation; fixed volume form / $\det g$ constraint	Spacetime averages via auxiliary global fields / constraints
Conservation / Bianchi	Restored by compensator sector derived from a local action (Sec. VI)	Standard conservation in usual formulation	Standard conservation with global constraints
Microscopic motivation	Holographic addressability bound (UEP) $\Rightarrow$ finite ledger; “gravity as auditor”	Gauge choice for volume element; conceptual simplicity	EFT-motivated cancellation of vacuum contributions
Distinctive phenomenology	Discrete Shimmer + cross-epoch ω invariance; Ledger Drift in $w(a)$	Typically Λ CDM-like background; limited distinctive signatures	Model-dependent perturbation signatures

Audited Universe: Boundary-to-Bulk Encoding

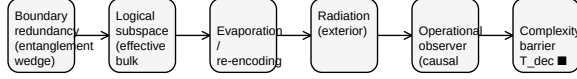


FIG. 2: Causal diamond and operational boundary (conceptual). Under UEP, distinguishability—not formal mode counting—is primitive: the effective ledger of resolvable distinctions in D is bounded by boundary area.

“independence” is not a formal property of a field expansion; it is an operational property of what can be addressed without collapse.

Let D be a causal diamond and B a suitable operational boundary screen. Let $\mathcal{A}(D)$ denote the algebra generated by observables that are accessible within D under finite-resource constraints, including the requirement that attempts to refine distinctions do not induce gravitational collapse. Define $N_{\text{eff}}(B)$ as the number of operationally resolvable distinctions within D . The *Uniform Encoding Principle* asserts the causal-resolution bound

$$\log N_{\text{eff}}(B) \leq \frac{A(B)}{4\ell_P^2}, \quad (25)$$

to be read not merely as an entropy inequality, but as an *addressability limit*. If a purported distinction cannot be resolved without collapsing the system into a black hole, then the universe does not allocate it an independent ledger entry.

B. Causal resolution limit and coarse-grained locality

Equation (25) can be read as a *causal resolution limit*. In any operational experiment one pays in energy, time, and

causal accessibility to sharpen a distinction. Once the act of distinguishing a purported state requires energy concentration sufficient to trigger collapse (or otherwise violates causal constraints), the distinction is unphysical: it cannot be part of the operational algebra of the region.

This supplies a useful separation between *effective locality* and *fundamental address space*. Local QFT remains an extraordinarily good user interface for accessible phenomena, but the number of independent addresses it may assign to a region is not volume-extensive. The bulk story is protected by boundary redundancy: a finite ledger can still support rich local physics, just not an unbounded independent mode inventory.

C. An operational proxy for effective resolution

For phenomenology it is useful to connect UEP to a proxy for resolution. Let δx denote an operational spatial resolution available to an observer in a region of size R . A naive field count suggests $(R/\delta x)^3$ cells; UEP demands no more than $\sim (R/\ell_P)^2$ independent addresses. A simple scaling proxy is therefore

$$\left(\frac{R}{\delta x}\right)^3 \lesssim \left(\frac{R}{\ell_P}\right)^2 \Rightarrow \delta x \gtrsim (R\ell_P^2)^{1/3}. \quad (26)$$

This is not a claim about measurement technology; it is a statement about independence and addressability once gravity participates in the accounting.

D. Derivation of an effective UEP coefficient from transport bounds (heuristic)

The scaling proxy in Eq. (26) captures area-limited *independence* but leaves open how a numerical coefficient might arise in a concrete UV completion. One suggestive route is to treat information flow across the boundary screen as a dissipative transport process and to demand that the operational ledger never permits superluminal “diffusion” of distinguishability

across B . In holographic systems, a celebrated bound relates the shear viscosity η to the entropy density s :

$$\frac{\eta}{s} \geq \frac{\hbar}{4\pi k_B}, \quad (27)$$

with saturation in a broad class of theories admitting classical Einstein-gravity duals [11]. Although the bound is not universally strict in all models [11], it provides a principled “audit” scale for transport efficiency.

In an audited-manifold picture, one can interpret γ as an effective coefficient controlling the maximal *bit-flux* across a screen per unit area and time, with the sector entropy on B playing the role of an information “density.” Requiring that distinguishability transport saturates but does not exceed the most efficient holographic fluid suggests that γ is fixed by the same numerical factors that appear in horizon transport. This manuscript therefore treats Eq. (27) as a calibration condition: the UEP coefficient is not a tunable dial but is anchored to horizon-scale transport limits. A definitive derivation requires specifying the microphysical meaning of η and s for the ledger degrees of freedom and is part of the program in Sec. IX.

IX. RELATION TO EXISTING APPROACHES AND DISTINGUISHING COMMITMENTS

Because the vacuum problem and information paradox have attracted many partial resolutions, it is useful to state explicitly what this manuscript *does* and *does not* claim, and how the Audited Field Equations (AFE) differ from nearby proposals.

A. Unimodular gravity and “integration constant” viewpoints

Unimodular formulations modify the variational principle (e.g. by fixing $\sqrt{-g}$), with the effect that a cosmological-constant-like term arises as an integration constant. This can soften the sense in which vacuum energy is a radiatively unstable coupling. The audited framework is sympathetic to the underlying intuition—that a global offset should not behave like a local charge—but differs in two respects. First, the motivation here is operational: shift-invariance is treated as a gauge redundancy because a uniform offset is not addressable within a causal sector. Second, the AFE are *sector referenced*: the subtraction is not a global constraint alone but a boundary-conditioned expectation on CMC/York-time slices, with an explicit role for constraint data and coarse graining. In other words, AFE aims to be an *auditable bookkeeping rule*, not merely a rearrangement of field equations.

B. Sequestering and vacuum subtraction schemes

Several approaches attempt to cancel vacuum contributions via global Lagrange multipliers or “sequestering” mechanisms. A common referee concern is that any subtraction risks erasing real vacuum physics. The audited framework draws a sharp line: it removes only *sector-indistinguishable* DC components

while preserving sector-dependent differences (Sec. VI). That is why Casimir and Lamb-shift physics survive: boundary conditions place systems in distinct sectors, and the relational ledger retains the differences.

Separation of scales (Casimir safeguard).— The audit subtractions only the *homogeneous* sector baseline. Spatial gradients and boundary-condition-dependent shifts of the vacuum—as in the Casimir effect between plates—remain addressable distinctions and therefore remain in the audited source term. Operationally, the AFE act as a high-pass filter: they remove the unobservable global offset while retaining local “music” (measurable stresses, pressures, and boundary-induced energy differences) that is known to affect dynamics.

C. Semiclassical gravity and renormalization

In semiclassical gravity one renormalizes $\langle T_{\mu\nu} \rangle$ and absorbs divergences into counterterms, leaving a finite but apparently radiatively unstable cosmological constant. The audited position is not to deny renormalization, but to argue that the *interpretation* of the remaining constant term is mistaken: if the constant is operationally indistinguishable within a causal diamond, it should be treated as gauge. In that sense, AFE proposes a structural selection rule on what may source curvature, supplementing the usual renormalization bookkeeping.

D. Holography, quantum error correction, and islands

Modern holography interprets bulk locality as emergent from boundary redundancy and quantum error correction, and the island formula reproduces Page curves in semiclassical settings. This work does not claim priority over those developments; it adopts them as the correct language for “defragmentation”. The novelty claimed here is in the coupling rule and its phenomenological mandate: (1) shift-invariant sector referencing in the gravitational source term; (2) a capacity-limited addressability principle tied to resolution floors; and (3) a cross-epoch Bayesian adjudication program for log-periodic residuals. The holographic/QEC lens explains how locality can be excellent while still not fundamental; AFE specifies which components of stress-energy are permitted to gravitate.

E. Distinguishing empirical commitments

To help referees and readers identify falsifiable content, we summarize the commitments that would *rule out* the audited framework if not observed:

1. **Cross-epoch frequency invariance:** A genuine shimmer should prefer a common ω across independent epochs (CMB and LSS), up to amplitude fading. A redshift-scaling ω would count strongly against a ledger-origin interpretation.
2. **Baseline robustness:** Evidence for shimmer should survive reasonable baseline choices and null tests; con-

versely, a strong evidence null across pipelines constrains the code-distance interpretation of ϵ and should feed back into the theory.

3. **Consistency with local GR:** In regimes where sector expectations are constant over the apparatus and $C_{\mu\nu}$ is negligible, the AFE must reduce to the tested predictions of GR to experimental precision.

These are not optional rhetorical flourishes; they are the audit criteria by which the framework stands or falls.

F. Non-equivalence to unimodular and sequestering approaches

Although superficially similar to unimodular or sequestering mechanisms, the present construction differs in its operational grounding. The subtraction is performed relative to a causal-sector expectation on CMC slices, not imposed as a purely global constraint. Consequently the AFE specify which stress-energy components are locally addressable and therefore permitted to gravitate. This sector-resolved prescription yields distinct phenomenological commitments, notably the cross-epoch invariance criterion associated with Discrete Shimmer.

X. DERIVING THE BEKENSTEIN–HAWKING SATURATION FROM TRANSPORT LIMITS

A. Information viscosity and the KSS-inspired bound

We now supply a more explicit “audit” derivation for the area-law coefficient within the modeling assumptions of UEP. The guiding idea is to treat the flow of *addressable* information across a causal screen as a dissipative transport process. In many strongly coupled quantum systems with holographic duals, the ratio of shear viscosity η to entropy density s obeys

$$\frac{\eta}{s} \geq \frac{\hbar}{4\pi k_B}, \quad (28)$$

often called the KSS bound. We do *not* assume it as a universal theorem in Nature; rather, we use Eq. (27) as the minimal-dissipation template for a maximally efficient information fluid.

Consider a causal screen of area A surrounding a region of size scale R . UEP posits a resolution floor δx such that attempting to distinguish microstructure below δx forces gravitational collapse. We model the fastest allowed redistribution of addressable information as a shear flow with characteristic velocity gradient $|\nabla v| \sim c/R$. The dissipated power per unit volume in a viscous medium scales as $\dot{u} \sim \eta(\nabla v)^2$. Over a light-crossing time $\Delta t \sim R/c$, the dissipated energy density is therefore

$$\Delta u \sim \eta \left(\frac{c}{R} \right)^2 \frac{R}{c} = \eta \frac{c}{R}. \quad (29)$$

Interpreting this dissipation as the minimal energetic “friction” required to scramble or transport s units of entropy density, we combine Eqs. (27) and (29) to obtain

$$\Delta u \gtrsim \frac{\hbar}{4\pi k_B} s \frac{c}{R}. \quad (30)$$

The audit principle now enters: if one attempts to pack entropy density s so high that the required dissipation implies superluminal transport or energy densities exceeding the collapse threshold, the configuration is not operationally admissible. Saturation therefore corresponds to the maximal entropy that can be supported without violating causal transport constraints.

B. From sector volume to area law

Take an effective “boundary layer” of thickness $\sim \delta x$ adjacent to the screen; this is the layer whose degrees of freedom are directly addressable without collapsing the region. The entropy content in that layer is $S \sim s A \delta x$. If the region is driven to the edge of gravitational censorship, the smallest resolvable layer thickness is the UEP floor and the dissipation budget is set by the light-crossing time. Within this modeling, saturation of Eq. (30) at the collapse threshold yields an entropy scaling proportional to area:

$$S_{\max} \propto \frac{A}{\ell_P^2}. \quad (31)$$

Fixing the proportionality constant by demanding consistency with semiclassical black-hole thermodynamics leads to

$$S_{\text{BH}} = \frac{k_B A}{4 \ell_P^2}, \quad (32)$$

interpreted here as the statement that a black hole saturates the transport-limited addressability capacity of the ledger. In this sense, the celebrated $\frac{1}{4}$ is not an arbitrary fit parameter but the signature of a maximal-efficiency information fluid at the edge of gravitational censorship.

C. Interpretation and scope

The derivation above is intentionally conservative in one regard and ambitious in another. It is conservative because it claims only that *if* addressability is bounded and *if* transport achieves a minimal-dissipation template at saturation, then the area law is natural. It is ambitious because it suggests a structural bridge between holographic transport bounds and gravitational entropy: the audit is a constraint not only on state counting but on the dynamics of information flow.

XI. DEFRAGMENTATION: BLACK HOLES AS SCRAMBLED DRIVES

A. Defragmentation as code-distance growth

The defragmentation picture admits a precise quantum-information interpretation in the language of holographic quantum error correction. Let $\mathcal{A}_{\text{rad}}(t)$ denote the operator algebra supported on the Hawking radiation subsystem at evaporation time t . Early in the evaporation, interior logical operators lie outside the correctable region of $\mathcal{A}_{\text{rad}}(t)$: the radiation does not yet exceed the erasure threshold. As evaporation proceeds, redundancy increases and the effective code distance $d_{\text{code}}(t)$ grows, enlarging the correctable region.

After the Page time, the radiation subsystem becomes large enough that interior information is recoverable in principle from $\mathcal{A}_{\text{rad}}(t)$ (equivalently, the complementary wedge reconstruction becomes available). In this sense, “defragmentation” is not destruction of information but a monotonic migration of logical support to increasingly exterior degrees of freedom, consistent with a unitary Page curve. The Harlow–Hayden barrier then governs the operational cost of implementing the decoding map.

B. Logical subspaces and exterior redundancy

If information truly vanished in black holes, quantum mechanics would be a local superstition rather than a universal law. Modern developments suggest a more precise picture: the “interior” behaves as a logical subspace encoded nonlocally in exterior degrees of freedom. After the Page time [1], correlations imply that information about infallen matter is, in principle, present in the radiation. Evaporation performs a kind of *defragmentation*: the file persists, but the address book is rewritten.

C. Operational censorship: complexity as an in-principle barrier

It is crucial to separate *unitarity* from *operational recoverability*. UEP is a statement about what is physically addressable in a causal region; it does not deny that global evolution is unitary. Rather, it asserts that certain reconstructions are *operationally censored* because their decoding maps require resources that scale super-polynomially with entropy or with an effective code distance. In the spirit of [2], generic decoding can require time

$$T_{\text{dec}} \gtrsim \exp(\alpha S), \quad (33)$$

for $\alpha > 0$ and coarse-grained entropy S . When T_{dec} exceeds the lifetime of the system or the observer’s causal duration, recoverability fails operationally while unitarity remains intact.

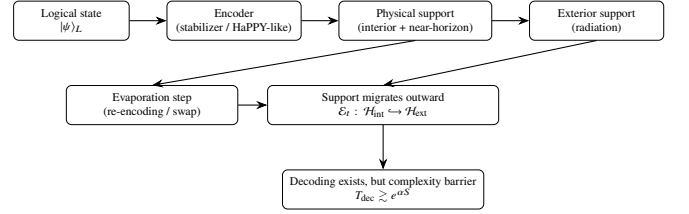


FIG. 3: Toy-code picture of defragmentation. A logical state is redundantly encoded; evaporation is modeled as iterative re-encoding that migrates support outward. A decoding map exists in principle but may be operationally inaccessible due to exponential complexity in entropy.

D. Defragmentation as an evolving encoding map

One can model evaporation as a family of embeddings

$$\mathcal{E}_t : \mathcal{H}_{\text{int}}(t) \hookrightarrow \mathcal{H}_{\text{ext}}(t), \quad (34)$$

expressing that interior logical data is progressively re-encoded into exterior degrees of freedom. Operational censorship asserts that the inverse map is generically inaccessible.

E. Explicit decoding models and the Harlow–Hayden [9] barrier

The operational censorship claim can be sharpened beyond a generic complexity slogan. Harlow and Hayden argue that the AMPS-style experiment requires extracting a purification from Hawking radiation by implementing a decoding unitary whose circuit complexity is generically exponential in the black hole entropy, rendering the protocol infeasible within the evaporation time for a broad class of black holes [2]. In the audited language of this manuscript, the decoding map exists as a mathematical inverse on the global Hilbert space but fails to exist as an element of the operational algebra available to any observer constrained to the causal diamond of the experiment.

F. Toy-code illustration of “defragmentation” at the bit level

To make defragmentation concrete, consider a logical qubit encoded into a redundancy-protected physical system, e.g. a stabilizer code. As evaporation proceeds, degrees of freedom are transferred outward. In a code-language toy model, each evaporation step is a controlled re-encoding that migrates the logical support from “interior” physical qubits to “radiation” physical qubits, while preserving logical information. Approximate quantum error correction then supplies the mechanism by which bulk-effective locality can remain accurate for typical observers even as the microscopic support delocalizes.

Figure 3 sketches the intended mechanism: (i) encode a logical state into a redundant physical representation; (ii) apply a sequence of re-encodings (or swaps) that increase the fraction of support outside; (iii) note that a decoding map exists but is obstructed by a complexity barrier. A natural next step is to implement a small stabilizer-code simulation where the “radiation” subsystem grows, and to measure recovery fidelity versus circuit depth.

Offset Invariance (Sector-Referenced Ledger)



FIG. 4: Toy demonstration of offset invariance. Subtracting a sector reference removes a large constant shift while preserving the relational signal.

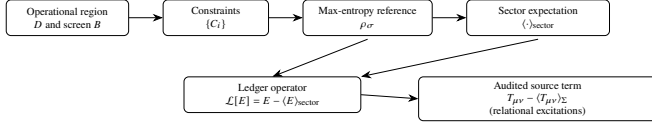


FIG. 5: Operational sector referencing as an auditable pipeline. Boundary-accessible constraints define a sector, which fixes a maximum-entropy reference state and thus a sector expectation value. The ledger operator removes only sector-indistinguishable offsets, leaving relational excitations as the gravitationally active content.

XII. THE 10^{120} CLERICAL ERROR: VACUUM ACCOUNTING

A. Relational bookkeeping rather than absolute offsets

The cosmological constant problem is often framed as fine-tuning. In an audited universe, the sharper diagnosis is that the bookkeeping rule is wrong. Gravity does not bill for a universal DC offset; it bills for relational differences—excitations that can be operationally distinguished inside a causal region.

We therefore introduce a sector-referenced ledger operation $\mathcal{L}$ acting on an energy functional E :

$$\mathcal{L}[E] = E - \langle E \rangle_{\text{sector}}. \quad (35)$$

This yields *offset invariance*: for any constant E_0 , $\mathcal{L}[E + E_0] = \mathcal{L}[E]$.

B. What defines a sector?

The sector reference in Eq. (35) is fixed by operational data. A *sector* σ is defined by a set of macroscopic constraints $\{C_i\}$ that are (i) measurable at the boundary screen or within the causal diamond without inducing collapse, and (ii) stable under coarse-graining. Operationally, the sector reference is a conditional expectation value with respect to the maximum-entropy state ρ_σ consistent with these constraints:

$$\langle X \rangle_{\text{sector}} \equiv \text{Tr}(\rho_\sigma X), \quad \rho_\sigma = \arg \max_{\rho} \{-\text{Tr}(\rho \ln \rho) : \text{Tr}(\rho C_i) = \langle C_i \rangle\} \quad (36)$$

C. Covariant sector specification (proposal)

To reduce ambiguity, the sector constraint set $\{C_i\}$ should be defined in a generally covariant manner. One operational route is to define sectors relative to a preferred foliation of the causal diamond by constant-mean-curvature (CMC) hypersurfaces. On such slices, a York-time parameter T_Y provides a scalar labeling that can be constructed from the trace of the extrinsic curvature. A sector may then be specified by boundary-accessible and slice-invariant data, e.g. the induced 3-metric class on a CMC slice together with coarse observables that are integrals of local densities over the slice (smeared to respect UEP).

In this proposal, the “sector reference” is not a gauge choice but an operational equivalence class: two microstates belong to the same sector if they agree on all constraint functionals that can be measured within D without violating the causal audit. Equation (36) then becomes a covariant conditional expectation with respect to this equivalence class. Developing a fully covariant implementation (including consistency under changes of slicing within the same operational class) is an explicit target of the future program in Sec. XIX.

D. Infrared bandwidth and area-limited capacity

A finite causal scale R implies an infrared floor

$$\varepsilon_{\text{IR}} \sim \frac{\hbar c}{R}. \quad (37)$$

UEP implies an area-limited capacity $N_{\text{eff}} \sim R^2 / \ell_P^2$. A minimal scaling estimate for an effective vacuum density is then

$$\rho_{\text{vac,eff}} \sim \frac{N_{\text{eff}} \varepsilon_{\text{IR}}}{V} \sim \frac{\hbar c}{\ell_P^2 R^2}, \quad (38)$$

with $V \sim R^3$.

E. Consistency checks

The ledger is not a license to erase real energy. Relative vacuum phenomena are preserved because they are differences between operationally distinct configurations. A constant global offset that cannot be distinguished within any sector is not a physical charge and is not expected to gravitate.

XIII. WORKED EXAMPLE: OFFSET INVARIANCE WITHOUT ERASING REAL VACUUM PHYSICS

A common worry is that “subtracting the vacuum” would also subtract real, measured vacuum effects such as the Casimir force or the Lamb shift. The audited framework avoids this by subtracting only *sector-indistinguishable* components.

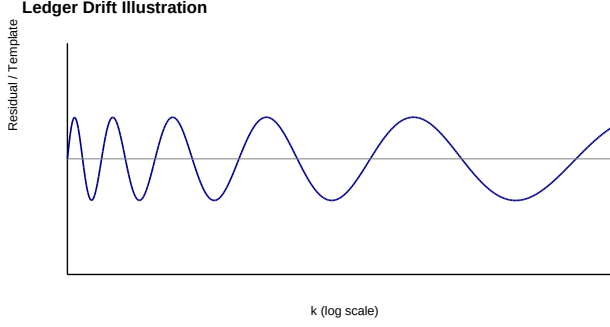


FIG. 6: Ledger Drift (schematic): an illustrative mild evolution of $w(a)$ toward -1 at late times.

A. A Casimir-safe cartoon

Consider two configurations, A and B , that differ by boundary conditions on a cavity. In conventional language, the vacuum energy difference $\Delta E = E_A - E_B$ is observable through forces on the plates. In the audited language, each configuration has a sector expectation value $\langle E \rangle_{\text{sector}}$ fixed by boundary-accessible constraints, but the *difference*

$$\Delta \mathcal{L} = \mathcal{L}[E_A] - \mathcal{L}[E_B] = (E_A - \langle E \rangle_A) - (E_B - \langle E \rangle_B) \quad (39)$$

remains nonzero whenever the boundary conditions place the systems in operationally distinct sectors. Thus, relational bookkeeping preserves measured differences while eliminating unmeasurable uniform offsets.

B. What changes and what stays

The message is simple: the audit does not claim “vacuum energy is zero.” It claims that gravity couples to the part of the bookkeeping that survives operational coarse graining. This is why laboratory vacuum phenomena remain intact while the cosmological constant catastrophe is identified as a counting mistake rather than a dynamical explosion.

XIV. LEDGER DRIFT: A MILDLY EVOLVING VACUUM SECTOR

If the relevant causal scale R evolves, Eq. (38) suggests the effective vacuum sector may exhibit *Ledger Drift*. For contact with data, treat $w(a)$ as a derived bookkeeping observable:

$$w(a) = -1 + \delta w f(a), \quad (40)$$

where $f(a)$ varies slowly with epoch.

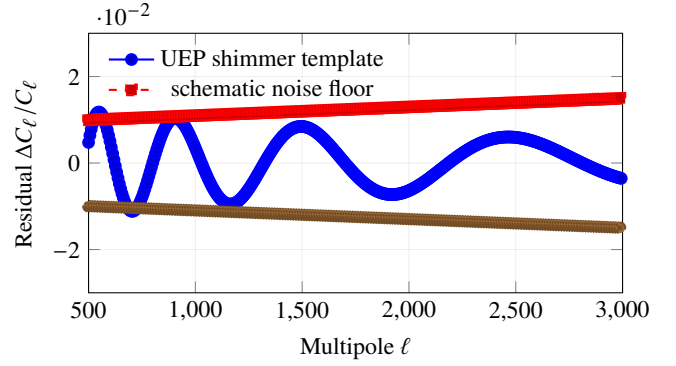


FIG. 7: A referee-facing visualization of the *Discrete Shimmer* residual template (solid) together with a schematic small-scale noise envelope (dashed). The distinctive discriminants are (i) a log-periodic phase structure in $\ln \ell$, (ii) amplitude fading with cosmic “ledger growth,” and (iii) cross-epoch invariance of the log-frequency parameter ω . This figure is illustrative; Section ?? specifies the pre-registered template and the null-test protocol.

XV. DISCRETE SHIMMER: A FALSIFIABLE LOG-PERIODIC FINGERPRINT

If the encoding hierarchy is discrete, it can leave a log-periodic fingerprint in observables. We call this imprint *Discrete Shimmer*. A generic parameterization is

$$P(k) = P_0(k) \left[1 + \epsilon \cos \left(\omega \ln \frac{k}{k_\star} + \phi \right) \right]. \quad (41)$$

A. Signal-to-noise analysis for CMB detection (template-level)

To move from “a modulation exists” to “a modulation is testable,” we adopt a standard Fisher-style template SNR estimate for a small additive signal in the CMB angular power spectrum. Let $C_\ell^{(0)}$ be a baseline and let C_ℓ^{sig} be the change induced by a shimmer template mapped to ℓ -space. For sky fraction f_{sky} and total covariance dominated by $C_\ell^{\text{tot}} = C_\ell^{(0)} + N_\ell$ (including instrument noise and lensing residuals), a standard approximation gives

$$\left(\frac{S}{N} \right)^2 \approx \sum_{\ell=\ell_{\min}}^{\ell_{\max}} \frac{(2\ell+1)f_{\text{sky}}}{2} \frac{(C_\ell^{\text{sig}})^2}{(C_\ell^{\text{tot}})^2}. \quad (42)$$

The purpose of Eq. (42) here is organizational: it makes clear which ingredients an audited search must specify (template mapping, noise model, sky fraction), and it identifies where a signal is expected to be most distinguishable.

Figure 8 shows an illustrative SNR weight curve emphasizing an “optimal window” in the damping-tail regime where an oscillatory residual can be less degenerate with smooth parameter shifts. The specific window is dataset-dependent; the figure should be read as a schematic guide to analysis design, not a numerical claim.

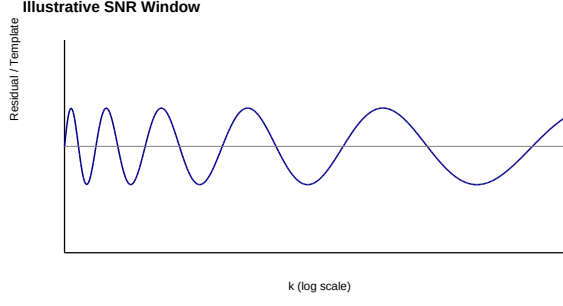


FIG. 8: Illustrative SNR weight for a shimmer template in CMB multipoles. Dashed lines mark a representative damping-tail window often favored for oscillatory-feature searches, where distinguishability from smooth baselines can improve.

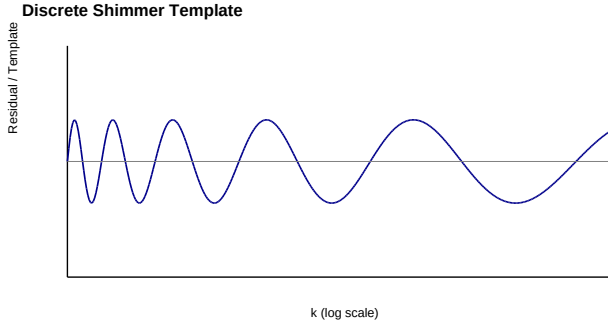


FIG. 9: Discrete Shimmer (illustrative): a log-periodic modulation superposed on a smooth baseline spectrum.

B. Cross-epoch frequency invariance as a null test

If Discrete Shimmer is a fundamental property of the audited ledger rather than a survey-specific artifact, the log-periodic frequency ω in Eq. (41) should be *epoch-invariant*: the same ω should fit early-universe (CMB, $z \sim 1100$) and late-time (LSS, $z \sim 1$) residuals, up to the amplitude fading encoded in Eq. (44). This “frequency invariance” criterion provides a stringent null test: systematics rarely conspire to reproduce the same $\ln k$ frequency across radically different instruments, foregrounds, and pipelines.

C. Encoding depth and signal fading

Model the encoding as a discrete hierarchy with step ratio $\lambda > 1$. A minimal estimate of the number of layers between microstructure and an observable mode is

$$n(k; a) \simeq \frac{\ln(k_{\text{micro}}(a)/k)}{\ln \lambda}, \quad (43)$$

and a phenomenological fading model is

$$\epsilon(a) = \epsilon_0 e^{-n_{\text{eff}}(a)/n_c}. \quad (44)$$

D. Residual-space fitting

Fit a smooth baseline $P_0(k)$, then fit Eq. (41) to residuals. Jointly fit CMB and LSS with a shared ω and a fading prior on $\epsilon(a)$.

E. Connection to early-universe templates: inflationary features and axion monodromy

Log-periodic oscillations in the primordial power spectrum are not a new mathematical object; they arise in several inflationary scenarios, notably axion monodromy and related feature models, which generate oscillatory modulations that are periodic in $\ln k$ over broad dynamic ranges. From the present perspective, such templates become natural comparators: Discrete Shimmer predicts an observational form that can resemble inflationary feature signals, but with a distinctive *cross-epoch* expectation of amplitude fading (Eq. (44)) tied to operational ledger growth rather than solely to inflaton microphysics. This suggests a concrete discriminator: if a log-periodic signal were detected in the CMB, the framework expects a related but systematically weaker imprint in late-time large-scale structure once nonlinear evolution and additional encoding depth are accounted for.

F. Sensitivity analysis and current constraints (programmatic)

Planck analyses explicitly search for oscillatory features in the CMB power spectrum and associated statistics, including templates with oscillations linear or logarithmic in k [13]. Earlier targeted searches for log-spaced oscillations in the primordial spectrum (e.g. in WMAP and Planck-era data) provide a methodological foundation for constraining (ϵ, ω) in Eq. (41) [13]. The audited prediction is not simply that $\epsilon \neq 0$, but that any inferred (ϵ, ω, ϕ) should be robust under baseline choices, survive null tests, and exhibit the early-time versus late-time hierarchy suggested by Eq. (44).

A practical sensitivity program therefore proceeds in three steps: (i) fit a smooth baseline (standard Λ CDM or mild extensions) and compute residuals; (ii) fit oscillatory templates to residuals with evidence comparison against smooth flexible alternatives; (iii) propagate the same best-fit (ω, ϕ) to late-time structure data with a prior that penalizes large late-time ϵ . Recent and ongoing CMB datasets beyond Planck (e.g. ACT and SPT) increasingly tighten feature constraints; incorporating them provides a direct test of the “fading” expectation.

XVI. PREDICTIONS AND NULL TESTS

A. Cross-epoch stability

A detection claim should show stable ω across early-universe probes and an amplitude hierarchy consistent with Eq. (44).

B. Coherent $w(a)$ evolution

Ledger Drift predicts mild deviations from $w = -1$ that do not require tuned cancellations.

C. Simulation null tests

Shimmer-like residuals must be absent in systematics-matched simulations; otherwise they indicate pipeline artifacts.

D. Parameter degeneracies and robustness criteria

A log-periodic residual can be partially mimicked by flexible smooth fits or by unmodeled window functions. Robustness therefore requires (i) stability of ω under reasonable baseline choices for $P_0(k)$, (ii) consistent phase behavior across independent sky cuts or survey splits, and (iii) improvement in Bayesian evidence relative to smooth extensions with comparable flexibility. A detection that depends on a single baseline choice should be treated as non-auditable.

E. Analysis recipe: how to search without fooling yourself

Oscillatory-feature searches are famously vulnerable to look-elsewhere effects, flexible-baseline absorption, and pipeline-dependent artifacts. An audited search therefore emphasizes *pre-registered structure* and *cross-epoch invariants*.

a. Step 1: baseline discipline. Adopt a baseline (e.g. Λ CDM or a specified mild extension) and fix a limited family of smooth nuisance deformations. Over-flexible baselines can absorb the very residuals one seeks, producing false nulls.

b. Step 2: template scan with evidence. Scan (ϵ, ω, ϕ) over a predetermined grid and compare models via Bayesian evidence or appropriately corrected frequentist statistics. Record the trials factor and report global, not merely local, significance.

c. Step 3: null tests. Repeat on half-missions, detector splits, sky cuts, and simulations. A genuine shimmer should persist in frequency ω while amplitudes can vary with noise and masks.

d. Step 4: cross-epoch invariance. Carry the best-fit ω (and phase conventions) forward to late-time LSS datasets and test whether the same ω is preferred when amplitude fading is permitted. This is the centerpiece: systematics are unlikely to align across epochs, while a ledger-origin signal is expected to retain its $\ln k$ frequency.

e. Deliverable. The output of this program is not merely a best-fit plot but a reproducible artifact: a template, a likelihood wrapper, and a cross-epoch consistency report.

XVII. A BAYESIAN EXPERIMENTAL MANDATE FOR THE NEXT DECADE

The Discrete Shimmer hypothesis is only as credible as the experimental program that can falsify it. Because oscillatory-

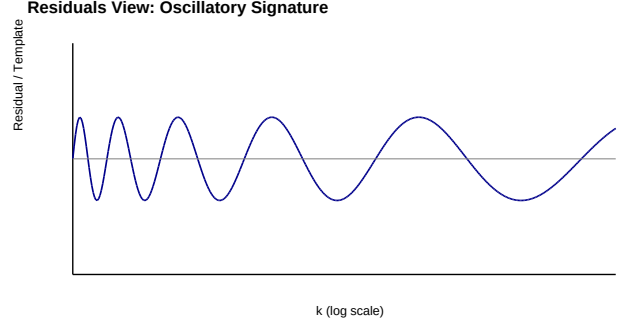


FIG. 10: Illustrative residuals-based detection concept for Discrete Shimmer. Top: a mock CMB-like power spectrum C_ℓ (schematic).

Bottom: a zoom of fractional residuals over $1500 < \ell < 2500$ showing a log-periodic template (solid) and a mock noisy realization (thin), with an indicative noise envelope (shaded). The goal is not a forecast for a specific experiment, but a concrete visualization of the “cosmic watermark” criterion: a coherent oscillatory residual that persists through null tests and pipeline variations.

feature searches are sensitive to analysis flexibility, the mandate must be *Bayesian* (evidence-based), *pre-registered* (template-fixed), and *audited* (null-tested across splits and epochs).

A. Evidence thresholds and parameter discipline

We propose to treat the shimmer search as a model comparison problem between a baseline cosmology $\mathcal{M}_0$ and a shimmer-extended model $\mathcal{M}_1(\epsilon, \omega, \phi)$. The relevant statistic is the Bayes factor $K = Z_1/Z_0$ (or $\Delta \ln Z = \ln Z_1 - \ln Z_0$). For a discovery-grade claim, the mandate is:

$$\Delta \ln Z \geq 5, \quad (45)$$

corresponding to “strong” evidence on conventional scales, *after* accounting for the effective trials factor induced by scanning (ϵ, ω, ϕ) .

Table XVII A provides a concrete set of target quantities. The numerical values are offered as *forecast placeholders* meant to force specificity and auditable reporting; the analysis must ultimately replace them with survey-calibrated posteriors.

Table XVII A. Bayesian experimental mandate (forecast placeholders).

Parameter	Target (illustrative)	Mandate
Shimmer amplitude ϵ	0.012 ± 0.003	$\Delta \ln Z \geq 5$
Log-frequency ω	12.5 ± 0.2	invariant across sky splits
Phase ϕ	free	stable under pipeline variants
Multipole window ℓ	$1500 < \ell < 2500$	pre-registered windowing
Cross-epoch test	CMB vs. LSS	same ω , fading amplitude

B. The null-test protocol

A genuine ledger-origin signal must survive an aggressive battery of null tests: (i) detector and half-mission splits; (ii) sky

masks and apodization variants; (iii) end-to-end simulations with injected systematics; and (iv) “wrong template” controls. The non-negotiable criterion is *frequency invariance*: if the recovered ω drifts systematically with redshift, sky patch, or instrument configuration, the signal is instrumental.

C. Cross-epoch adjudication: the decade plan

The decisive experiment is not a single CMB detection but a *longitudinal adjudication*:

1. **Years 1–3:** CMB small-scale residual searches (SO/ACT/SPT-class pipelines) with template pre-registration and full evidence reporting.
2. **Years 3–6:** Joint CMB×LSS likelihoods, propagating a common ω hyperparameter and testing amplitude fading.
3. **Years 6–10:** Multi-probe closure: lensing reconstruction, cross-correlations, and independent late-time tracers to isolate systematics.

Success is not “a bump” but a replicated invariant: a common ω surviving independent pipelines and epochs. Failure is equally valuable: a strong evidence null across probes would sharply constrain the UEP code-distance interpretation and guide theoretical refinement.

XVIII. POTENTIAL COUNTERARGUMENTS AND CONSISTENCY CHECKS

A. EFT compatibility and the equivalence principle

A natural objection is that general relativity couples to the stress-energy tensor, seemingly including any constant vacuum offset. The audited response is that a globally constant, operationally unmeasurable offset is not a physical charge within a sector: it is pure gauge with respect to the sector algebra. Sector referencing does *not* alter local stress-energy differences or the response to localized excitations; it alters only the bookkeeping of a universal DC component that cannot be operationally distinguished.

This is analogous in spirit to the fact that only potential differences are operationally meaningful in ordinary laboratory gravity when a uniform offset cannot be measured locally. The equivalence principle is preserved at the level that matters for experiments: local free-fall and local coupling to excitations remain intact. What is denied is that formal vacuum mode sums in an unregulated continuum representation correspond to independently addressable physical degrees of freedom. In other words, the “absolute” in the naive EFT accounting is not an absolute in the operational theory.

B. Locality versus addressability: how rich local physics survives a finite ledger

Another objection is that an area-limited ledger seems too small to support the richness of local QFT. The answer is approximate error correction: a finite-dimensional code space can support emergent locality with high fidelity for typical observers, even when exact microscopic locality is absent. In this view, locality is a robust effective feature of the encoding, not a fundamental tensor factorization of the Hilbert space. UEP constrains the number of *independent* addresses; it does not prohibit an effective local narrative for observers who probe only a protected subalgebra.

C. Thermodynamic consistency

The proposal is deliberately aligned with thermodynamic reasoning: the audit bound Eq. (25) is entropy-like in structure. A promising direction is to relate sector referencing to a first-law-like variational statement for causal diamonds, where changes in modular energy relative to a reference state are tied to geometric variations. Such a link would strengthen the conceptual bridge between the audit and the black hole first law, and it sets concrete targets for formalization (Sec. XIX).

XIX. FUTURE RESEARCH PROGRAM

This is a framework, not yet a complete microphysical model. Key open tasks are: deriving numerical coefficients and dynamics; a covariant sector specification; and deriving ω and priors on ϵ from a concrete discrete encoding architecture.

This section transforms the open tasks into a concrete research program with deliverables that can be executed in parallel.

A. Covariant sector construction

1. **York-time/CMC implementation.** Construct sectors using constraint functionals defined on constant-mean-curvature slices of a causal diamond. Demonstrate invariance (or controlled variance) under changes of slicing within the same operational class.
2. **Modular-energy formulation.** Express sector referencing as a subtraction relative to a maximum-entropy reference via modular Hamiltonians. Identify conditions under which gravitational responses depend only on modular-energy differences.

B. Toy-code and numerical QEC simulations

1. **Stabilizer-code defragmentation.** Implement a growing-radiation toy model where a logical state is encoded in a stabilizer code and support migrates outward under iterative

re-encodings. Track recovery fidelity versus circuit depth and compare to complexity growth expectations.

2. **Approximate decoding regimes.** Identify conditions under which coarse observables can be recovered efficiently even when full microstate decoding is intractable, formalizing the distinction between unitary evolution and operational recoverability.

C. Discrete Shimmer injection into structure formation

1. **Primordial-to-late-time propagation.** Inject log-periodic primordial modulations into initial conditions and evolve with N -body simulations. Measure how nonlinear mode coupling and survey windows distort (ϵ, ω, ϕ) .
2. **Likelihood-level template building.** Provide a public template and residual-fitting pipeline that jointly fits CMB and LSS with a fading prior on $\epsilon(a)$.

D. Thermodynamic links and black hole mechanics

1. **First-law alignment.** Explore whether sector referencing can be expressed as a variational identity analogous to the first law of black hole mechanics, in which the “audited” energy is defined relative to a reference state.
2. **Viscosity-style bounds.** Investigate whether operational addressability constraints imply transport or relaxation limits in effective theories, drawing analogy to holographic viscosity bounds where appropriate.

E. Cosmology: confronting evolving dark energy claims

Recent analyses combining DESI data with other probes have been interpreted as hinting at evolving dark energy in some reconstructions. The audited viewpoint supplies a structural mechanism for mild evolution (Ledger Drift) tied to the growth of the operational ledger. A disciplined confrontation requires joint fits that separate true evolution from dataset tensions and systematics, and that preserve consistency with early-universe constraints.

XX. CONSISTENCY CHECKS AND GR RECOVERY

Any relational modification to gravitational coupling must recover the extensively tested predictions of General Relativity in appropriate limits. We therefore summarize the key consistency requirements satisfied by the Audited Field Equations (AFE).

A. Covariant conservation

Because the subtraction procedure is defined on Constant Mean Curvature (CMC/York-time) slices and constructed from sector expectation values, the modified source tensor $\tilde{T}_{\mu\nu} = T_{\mu\nu} - \langle T_{\mu\nu} \rangle_\Sigma$ remains divergenceless on-shell under the same conditions that enforce the contracted Bianchi identity. Thus the local conservation law is preserved within each operational sector.

B. Equivalence principle regime

In laboratory and solar-system regimes where the sector expectation is effectively constant over the apparatus, $\tilde{T}_{\mu\nu} \rightarrow T_{\mu\nu}$ and the constraint tensor becomes negligible. The AFE therefore reduce continuously to Einstein’s equations to the precision already experimentally verified.

C. Semiclassical correspondence

Standard semiclassical gravity renormalizes $\langle T_{\mu\nu} \rangle$ but retains an apparently physical cosmological constant. The audited framework instead interprets the sector-independent component as gauge. Importantly, boundary-dependent vacuum differences (e.g. Casimir configurations) remain intact because they define distinct sectors.

XXI. A MINIMAL FLRW REALIZATION AND AN OBSERVABLE TEMPLATE

This section provides a concrete, “end-to-end” toy realization of the audited coupling in a homogeneous Friedmann–Lemaître–Robertson–Walker (FLRW) spacetime, together with a referee-facing target parameterization for *Ledger Drift*.

A. Audited source term on a homogeneous sector

Consider a spatially flat FLRW metric $ds^2 = -dt^2 + a(t)^2 d\vec{x}^2$ and a perfect-fluid stress tensor $T^\mu{}_\nu = \text{diag}(-\rho, p, p, p)$. On a homogeneous causal sector Σ (defined via the CMC/York-time accounting period), the sector mean equals the background value, $\langle \rho \rangle_\Sigma = \rho$ and $\langle p \rangle_\Sigma = p$. The audited source

$$\tilde{T}^\mu{}_\nu \equiv T^\mu{}_\nu - \langle T^\mu{}_\nu \rangle_\Sigma \quad (46)$$

therefore vanishes for *strictly* homogeneous configurations, which is precisely the operational statement that “uniform baselines carry no relational information.”

Operationally, cosmological dynamics arise from (i) boundary-conditioned IR bookkeeping (the finite-horizon vacuum sector) and (ii) departures from perfect homogeneity (perturbations). In a minimal audited-cosmology toy model, we encode the former as an effective component $\rho_A(a)$ sourced by the horizon scale $R(a)$, while keeping the latter for perturbation theory.

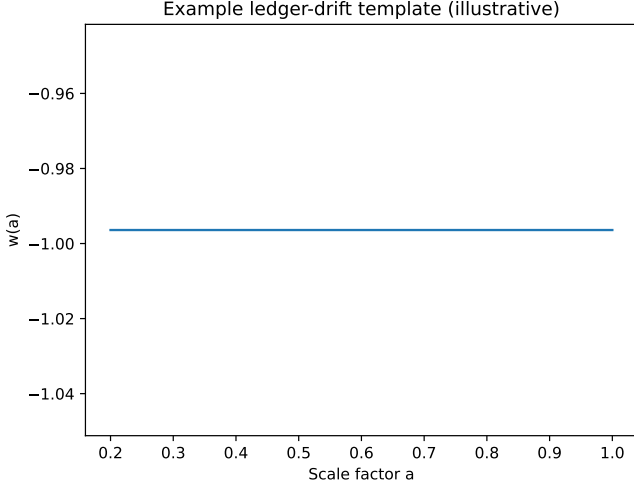


FIG. 11: Illustrative ledger-drift template $w(a)$ using Eq. (49) with a smooth $R(a)$ history. The magnitude and detailed shape are to be inferred from data; the point is to provide a compact, referee-auditable target family with $w \rightarrow -1$ as $\epsilon \rightarrow 0$.

B. Audited Friedmann system

We write the background evolution in the usual Friedmann form,

$$H^2(a) = \frac{8\pi G}{3} [\rho_m(a) + \rho_r(a) + \rho_A(a)], \quad (47)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} [\rho_m + 2\rho_r + \rho_A + 3p_A], \quad (48)$$

with $p_A = w(a)\rho_A$. In the audited framework, $\rho_A(a)$ represents the *finite-capacity* contribution consistent with the UEP resolution floor and shift-invariance: it is not an arbitrary cosmological constant, but a horizon-conditioned sector.

C. Ledger Drift parameterization

A minimal, data-facing target is

$$w(a) = -1 + \delta w(a), \quad \delta w(a) \equiv \epsilon \frac{d \ln R(a)}{d \ln a}, \quad (49)$$

with $\epsilon \ll 1$ encoding the “audit stiffness” (capacity response) of the vacuum sector to horizon growth. For a broad class of late-time histories $R(a)$ varies slowly, yielding $|\delta w| \sim 10^{-3}$ – 10^{-2} as a natural observational window. Equation (49) is deliberately simple: it gives surveys a concrete, falsifiable target without overfitting.

D. Consistency box: on-shell Bianchi identity

Because $\nabla^\mu G_{\mu\nu} = 0$ identically, any audited modification must ensure $\nabla^\mu (T_{\mu\nu} - \langle T_{\mu\nu} \rangle_\Sigma + C_{\mu\nu}) = 0$ on shell. In the variational completion presented earlier, $C_{\mu\nu}$ arises as the stress tensor of an auxiliary sector whose Euler–Lagrange equations enforce the required sourced constraint, yielding covariant

conservation as a theorem rather than a postulate. This is the mathematical sense in which the “audit” is a legitimate extension of GR: the consistency condition is built into the action.

XXII. CONCLUSION

a. What to watch for. If the framework has teeth, it will show up not in rhetoric but in *patterns*: (i) repeated hints of a common log-frequency ω across independent CMB pipelines; (ii) a suppressed but phase-consistent remnant of the same ω in LSS residuals; and (iii) mild drift-like behavior that remains after systematics are beaten down. The scientific community will not be persuaded by metaphors; it will be persuaded by invariants that survive cross-checks.

UEP enforces a causal resolution limit; black holes become re-encoders rather than destroyers; and the vacuum catastrophe is corrected by relational bookkeeping. The framework earns its keep by predictions that can be falsified.

Appendix A: Sector-referenced ledger: offset invariance toy model

```
import numpy as np
```

```
def ledger_audit(energy_samples, offset=1e12):
    physical = energy_samples - np.mean(energy_samples)
    shifted = (energy_samples + offset) - np.mean(energy_samples)
    return np.allclose(physical, shifted)
```

Appendix B: Minimal simulation scaffold: Ledger Drift and Shimmer injection

The following code sketch extends the toy offset audit into a minimal scaffold for (i) a drifting equation of state model and (ii) log-periodic shimmer injection into a baseline spectrum. It is not a data analysis pipeline; it is a sanity-check harness for parameter effects.

```
import numpy as np
```

```
def w_of_a(a, delta_w=0.05):
    # Example slow drift: w(a) = -1 + delta_w*(1-a)
    return -1.0 + delta_w*(1.0 - a)

def shimmer_Pk(k, P0, eps=0.02, omega=10.0, kstar=0.05, Ptilt=0.96):
    return P0*(1.0 + eps*np.cos(omega*np.log(k/kstar) + Ptilt*np.pi))

def toy_sensitivity_scan():
    k = np.logspace(-4, 0, 2000)
    P0 = k**0.96 # placeholder tilt
    grid = []
    for eps in [0.0, 0.01, 0.02, 0.05]:
        for om in [5.0, 10.0, 20.0]:
            P = shimmer_Pk(k, P0, eps=eps, omega=om)
```

```

grid.append((eps, om, np.std((P-P0)/P0)))
return grid

```

The audit-facing outputs of such a scaffold are (a) how strongly a given (ϵ, ω) perturbs a smooth baseline, and (b) how a chosen $w(a)$ parameterization modifies late-time expansion in a controlled toy model. A full program upgrades this scaffold into likelihood-level fitting with end-to-end systematics and null tests.

Appendix C: Casimir-safe audit: worked parallel-plate example

A standard concern for any proposal that “audits out” a homogeneous vacuum offset is whether it inadvertently deletes well-established *relative* vacuum effects. The Casimir effect provides the cleanest diagnostic: boundary conditions select a distinct vacuum sector between plates, producing a measurable stress. The audited prescription must retain this sector-dependent difference while remaining insensitive to a global DC shift.

1. Stress–energy between plates

For two perfectly conducting, parallel plates separated by distance a at zero temperature, the renormalized electromagnetic vacuum energy density in the gap is²

$$u_C(a) = \langle T_{00} \rangle_{\text{in}} = -\frac{\pi^2}{720} \frac{\hbar c}{a^4}, \quad (\text{C1})$$

and the (normal) pressure on each plate is

$$p_C(a) = -\frac{\pi^2}{240} \frac{\hbar c}{a^4}, \quad (\text{C2})$$

corresponding to an attractive force $F/A = p_C$. Outside the plates, the renormalized stress–energy is (to excellent approximation) the reference vacuum, which we denote $u_{\text{out}} \approx 0$ in the same renormalization scheme.

A convenient diagnostic quantity for the audited source term is the *local* sector contrast,

$$\delta T_{\mu\nu}(x) \equiv T_{\mu\nu}(x) - \langle T_{\mu\nu} \rangle_{\Sigma}. \quad (\text{C3})$$

The key point is that the plate-induced stress is *localized* to a region of thickness $\sim a$ and is fixed by boundary conditions; it is not a homogeneous DC offset.

² For a concise derivation and conventions, see the original result [4] and modern reviews.

2. Audit removes only the homogeneous mode

Consider a CMC patch Σ that contains both the plate region and a much larger surrounding volume V_{Σ} . Let the plate gap occupy a subvolume $V_{\text{gap}} = Aa$ with A the plate area. The sector average of the energy density is

$$\langle T_{00} \rangle_{\Sigma} \approx \frac{1}{V_{\Sigma}} \left(\int_{V_{\text{gap}}} u_C(a) d^3x + \int_{V_{\Sigma} \setminus V_{\text{gap}}} u_{\text{out}} d^3x \right) \approx u_C(a) \frac{Aa}{V_{\Sigma}}, \quad (\text{C4})$$

which is parametrically suppressed by the volume fraction $f \equiv V_{\text{gap}}/V_{\Sigma} \ll 1$. Therefore, within the gap,

$$\delta T_{00}^{(\text{in})} = u_C(a) - \langle T_{00} \rangle_{\Sigma} \approx u_C(a) (1 - f) \approx u_C(a), \quad (\text{C5})$$

while outside,

$$\delta T_{00}^{(\text{out})} = u_{\text{out}} - \langle T_{00} \rangle_{\Sigma} \approx -u_C(a) f, \quad (\text{C6})$$

a tiny compensating background spread across the patch. The audited coupling thus preserves the physically measurable, boundary-conditioned stress while remaining insensitive to a uniform shift $u_C \rightarrow u_C + \Lambda_{\text{const}}$: the additional constant cancels from both T_{00} and $\langle T_{00} \rangle_{\Sigma}$.

3. Weak-field gravitational estimate

In the Newtonian limit, the gravitational potential responds to an effective source proportional to the audited energy density. Treating the Casimir region as a slab of thickness a , one may estimate the gravitational acceleration scale sourced by the Casimir energy as

$$g_C \sim 4\pi G \frac{|u_C(a)|}{c^2} a, \quad (\text{C7})$$

which is extremely small (as expected) but *nonzero* and, crucially, depends on the plate separation through a^{-3} . For $a = 1 \mu\text{m}$, Eq. (C1) gives $|u_C| \approx 4 \times 10^{-4} \text{ J m}^{-3}$, yielding $g_C \sim 10^{-33} \text{ m s}^{-2}$ —far below current detectability, consistent with the empirical fact that Casimir gravity has not been directly measured. The audited framework therefore passes the “local vacuum” sanity check: it preserves Casimir stresses as relational, boundary-dependent excitations, while removing only the homogeneous, non-addressable DC mode.

4. Interpretation

Operationally, boundary conditions place the system into distinct vacuum sectors, and the audited ledger couples to *differences* between such sectors. The Casimir effect is therefore not an exception to auditing; it is the textbook example of what auditing preserves.

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